

- **Deleted carboxysome compartment (cx)**

- **Deleted metabolites relevant to compartment cx**

- 1- meta_M_h2o_cx
- 2- meta_M_2pglyc_cx
- 3- meta_M_h_cx
- 4- meta_M_rb15bp_cx
- 5- meta_M_3pg_cx
- 6- meta_M_hco3_cx
- 7- meta_M_co2_cx
- 8- meta_M_o2_cx

- **Deleted reactions:**

- 1- meta_R_O2tcx
- 2- meta_R_RB15BPtcx
- 3- meta_R_3PGtcx
- 4- meta_R_HCO3tcx
- 5- meta_R_2PGLYCtcx
- 6- meta_R_HCO3E_1_cx moved to cytosol
- 7- meta_R_RBPCcx moved to cytosol
- 8- meta_R_H2Otcx
- 9- meta_R_Htcx

- **Deleted parameters**

The medium was directly defined in the model by setting the corresponding parameter values. Therefore, this definition was removed from the model so that it can be used with our defined medium.

- **Directionality change**

The direction of following reactions was changed to match with the map.

- 1- R_EX_no3 should be irreversible due to the map.
- 2- R_NO3tex should be irreversible due to the map.
- 3- Direction of cit_c -> icit_c was corrected due to the map.
- 4- icit_c -> akg should be irreversible due to the map.
- 5- Direction R_ASPTA was corrected due to the map.
- 6- Direction of PGK was corrected due to the map.
- 7- Direction PSERT was corrected due to the map.
- 8- Direction ACKr was corrected due to the map.
- 9- Direction Acs was corrected due to the map.
- 10- Direction PPS was corrected due to the map.
- 11- Direction R_GLYCL was corrected and nh4 was replaced by nh3 due to Brenda and uniprot.

- **Further changes**

- 1- Reaction GLYCTO1 was added and GLYCLTDx was removed. Both are for glycolate to glyoxylate. However the removed one is the cost of nadh.
- 2- The demand reaction R_DM_akg_c had no BiGG ID, so instead of that there is SK_akg_c. Therefore it was corrected and direction was also corrected.

3- The demand reaction R_DM_fum_c had no BiGG ID, so instead of that there is SK_fum_c. Therefore it was corrected and direction was also corrected.

4- ASADi was old identifier and was changed to ASAD

- **Reactions and Metabolites newly added**

reactions_to_add

ID	Name	BiGG ID	Gene	Source
PDHcr	Pyruvate dehydrogenase (dihydrolipoamic	RXN-18331	phdD	Map evidence
PDHbr	Pyruvate dehydrogenase (dihydrolipoamic	DIHYDLIPACETRAI	PdhC	Map evidence
PdhB	Pyruvate dehydrogenase (lipoamide)	-	PdhB	Map evidence. No BiGG ID, but
SUCDi	Succinate dehydrogenase (irreversible)	SUCDi	SdhC	Map evidence
SGAT	Serine--glyoxylate aminotransferase	SGAT	Sgat	Map evidence
BHCC	3-hydroxy-D-aspartate aldolase	-	BhcC	New Pathway. No GiGG ID, but
BHCB	(3S)-3-hydroxy-D-aspartate dehydratase	-	BhcB	New Pathway. No GiGG ID, but
BHCD	iminosuccinate reductase	-	BhcD	New Pathway. No GiGG ID, but
BHCA	glycine--oxaloacetate transaminase	-	BhcA	New Pathway. No GiGG ID, but
RBPC	Ribulose-bisphosphate carboxylase	RBPC	Rubisco	Map evidence. Carboxylation
RBCh	Ribulose-bisphosphate carboxylase	RBCh	Rubisco	Map evidence. Oxygenation
HCO3E	HCO ₃ equilibration reaction	HCO3E	ca, chpx	Map evidence
NH3c	Ammonium proton dissociation	NH3c	Experimental s	Since reactions are based on A

metabolites_to_add

ID	Name	BiGG ID	Source
ehasp	betaHydroxyaspartate	-	Belong to New Pathway
isucc	iminosuccinate	-	Belong to New Pathway